STAFF REVIEW · H446 043 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q2; Booklet Q3 = original online Q3

Do Now — source answers

1. Allows access to and modification of caller variable

2. Where a variable is accessible

3. No value

4. Fast average access when well designed

I Do — source worked steps

1. Pause and inspect — Call Sum with [4,6]. Step Into enters the method; Step Over a call would run it without following its internal statements. Watch total=0, Length=2 and i=0.

2. Locate the first invalid access — At i=0 total becomes 4; at i=1 it becomes 10. The faulty test allows i=2, but valid indexes are 0 and 1. The next array access fails.

3. Correct and retest — Change i<=marks.Length to i<marks.Length. Retest [4,6]→10, []→0 and [7]→7. A small routine can now be tested independently and reused.

We Do — guided CHECK

1. Check your thinking
Not all code paths return a value.

2. Discuss, then respond
Why step into a function in a debugger?
Check your thinking
To inspect statements inside the called method.

You Do — source mark schemes

1. Reuse and easier isolated testing/maintenance.

2. Pauses execution to inspect current state and follow iteration.

3. Replace Console.WriteLine with return a+b;

Exit Ticket — source answer

1. True. Modular code separates responsibilities, making individual modules easier to test.